biology with mathematics. It must be one or the other," the headmaster proclaimed. Mathematics was more precious to Roger than biology. "All right, I'll do mathematics, chemistry, and physics," he said. When Roger got home that evening his parents were furious. They accused Roger of keeping bad company. Biology was essential to a medical career; how could he give it up?

Two years later came the decision of what to study in college. "I proposed to go to University College, London, and study for a mathematics degree," Roger recalls. "My father didn't approve at all. Mathematics might be all right for people who couldn't do anything else, but it wasn't the right thing to make a real career of." Roger was insistent, so his father arranged for one of the College's mathematicians to give him a special test. The mathematician invited Roger to take all day on the test, and warned him that he probably would be able to solve only one or two of the problems. When Roger solved all twelve problems correctly in a few hours, his father capitulated. Roger could study mathematics.

Roger initially had no intention of applying his mathematics to physics. It was pure math that interested him. But he got seduced.

The seduction began in 1952, when Roger as a fourth-year university student in London listened to a series of radio talks on cosmology by Fred Hoyle. The talks were fascinating, stimulating—and a bit confusing. A few of the things Hoyle said didn't quite make sense. One day Roger took the train up to Cambridge to visit his brother Oliver, who was studying physics there. At the end of the day, over dinner at the Kingswood restaurant, Roger discovered that Dennis Sciama, Oliver's officemate, was studying the Bondi–Gold–Hoyle steady-state theory of the Universe. How wonderful! Maybe Sciama could resolve Roger's confusion. "Hoyle says that according to the steady-state theory the expansion of the Universe will drive a distant galaxy out of sight; the galaxy will move out of the observable part of our Universe. But I don't see how this can be so." Roger pulled out a pen and began drawing a spacetime diagram on a napkin. "This diagram makes me, think that the galaxy will become dimmer and dimmer, redder and redder, but will never quite disappear. What am I doing wrong?"

Sciama was taken aback. Never had he seen such power in a space time diagram. Penrose was right; Hoyle had to be wrong. More important, Oliver's little brother was phenomenal.